OpenClaw Three Dialogue Solution Configuration Tests

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1. Course Overview

This course will introduce three common dialogue methods in OpenClaw, and provide their respective configuration and testing methods:

- whatsapp dialogue
- Local TUI dialogue
- AI voice interaction dialogue

After completing this course, you will be able to choose the most suitable dialogue method for your scenario and complete basic verification.

Note: whatsapp robot and AI voice interaction use the default model. If you need to change the default large model, see [03. OpenClaw Model Switching]

2. Three Dialogue Solutions Introduction

2.1 whatsapp Solution

Suitable scenarios:

- Mobile phone remote control
- Let multiple people experience OpenClaw together

Features:

- No need to stay by the device all the time
- Suitable for demonstrating "remote natural language control"

2.2 TUI Solution

Suitable scenarios:

- Local debugging
- Quick testing
- Troubleshooting model and skill invocation issues

Features:

- Most direct
- Easiest to see commands and return results

2.3 Voice Interaction Solution

Suitable scenarios:

- Complete AI interaction experience
- Voice wake-up and voice control
- Desktop assistant-style experience

Features:

- Strongest sense of interaction

3. whatsapp Dialogue Configuration and Testing

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

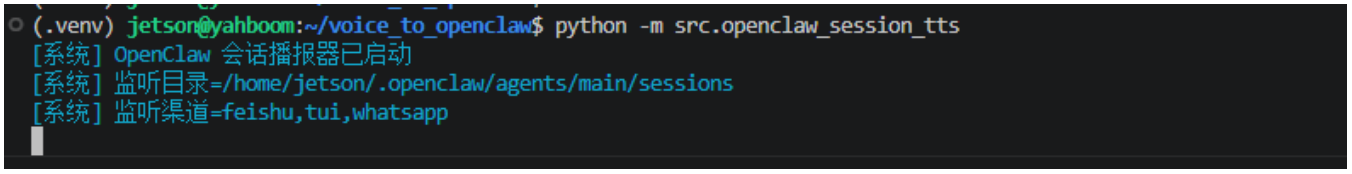
```
ENABLE_TTS=true
TTS_ENGINE=xunfei_ws
TTS_API_URL=wss://tts-api.xfyun.cn/v2/tts
TTS_VOICE=x4_xiaoyan
TTS_SAMPLE_RATE=16000
TTS_VOLUME=50
TTS_TIMEOUT=60
XUNFEI_TTS_SPEED=50
XUNFEI_TTS_PITCH=50
XUNFEI_TTS_TEXT_ENCODING=UTF8
XUNFEI_TTS_SIGN_HOST=ws-api.xfyun.cn
WAKE_ACK_TEXT=I am here.
```

```
TTS_MAX_CHARS=200
TTS_TRUNCATE_MODE=smart
TTS_LOCK_PATH=/tmp/voice_to_openclaw_tts.lock
# Terminal log language: zh / en
UI_LANGUAGE=en
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

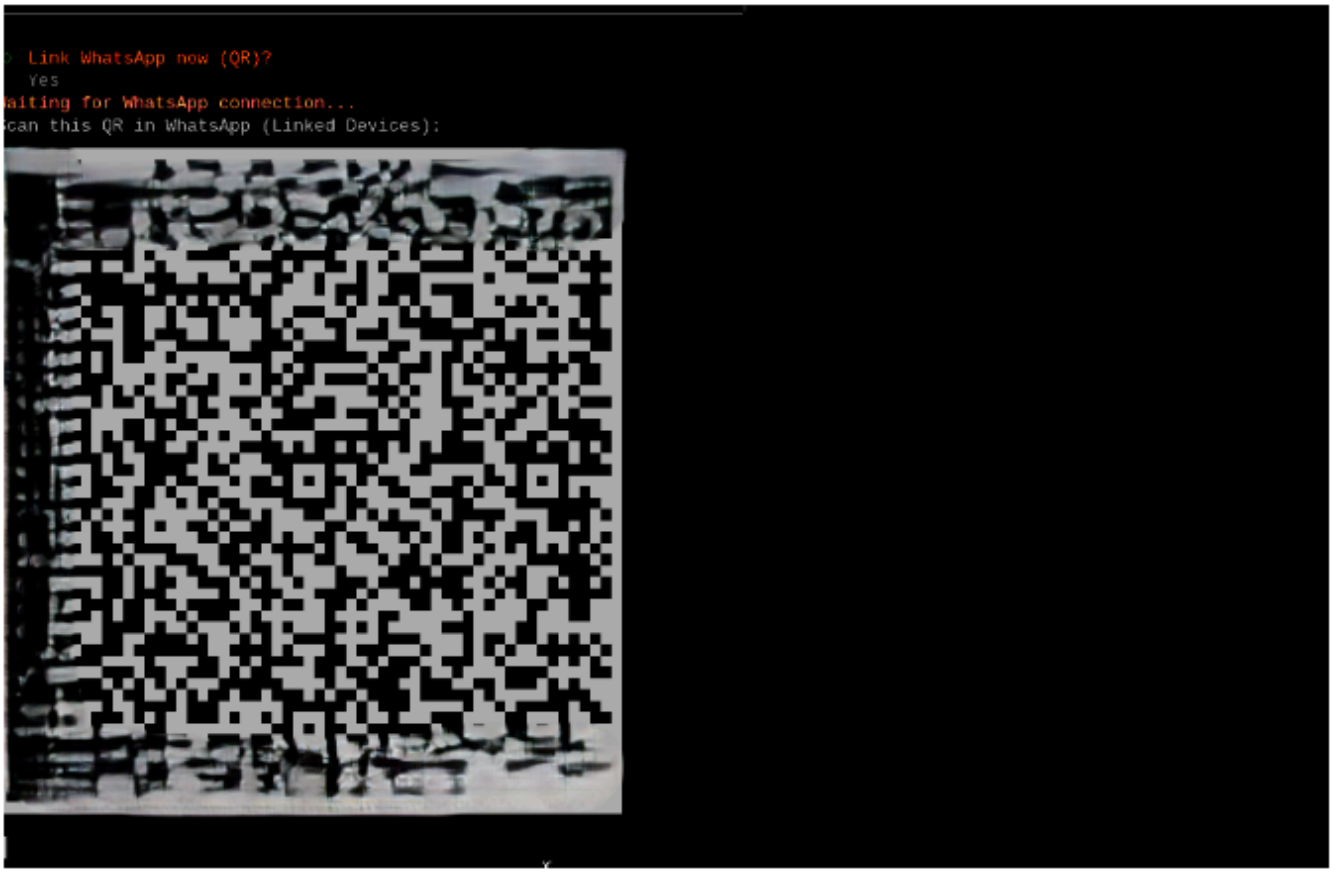


3.1 Install whatsapp Plugin

Terminal input:

```
openclaw channels login --channel whatsapp
```

After execution, a QR code will usually appear.



- Scan the QR code on the mobile phone to connect. "Linked after restart" appears, proving the connection is successful

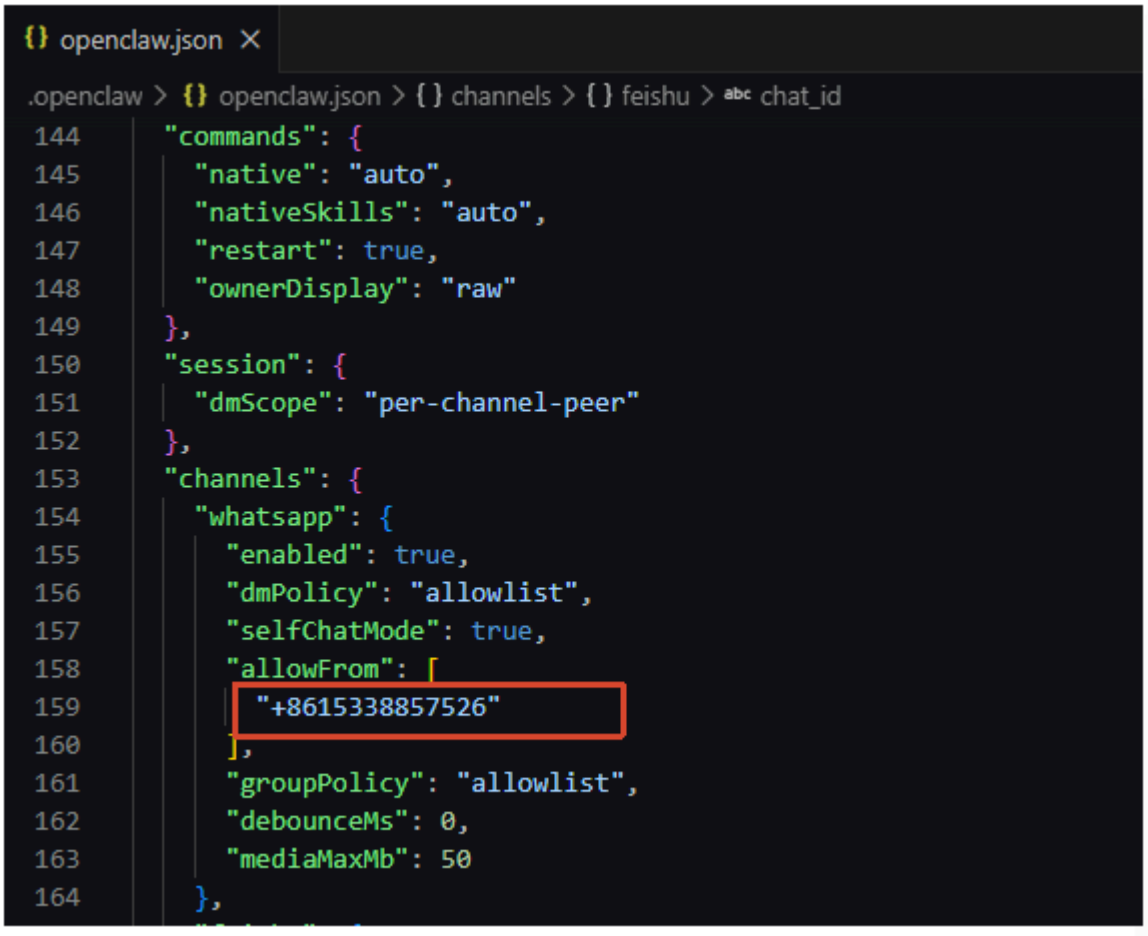
```
WhatsApp asked for a restart after pairing (code 515); creds are saved. Restarting connection once…
✅ Linked after restart; web session ready.
```

- Open the newly created conversation to see the conversation ID



- Fill this into "allowFrom" in the openclaw.json file:

```
nano /home/jetson/.openclaw/openclaw.json
```



- Restart OpenClaw gateway

```
openclaw gateway restart
```

3.2 Testing Method

After creation, you can directly send test messages in whatsapp, for example:

- Tell me a joke



3.3 Common Problems

- Cannot create robot after scanning code: First check network
- Created successfully but no reply: Check if OpenClaw gateway has restarted successfully

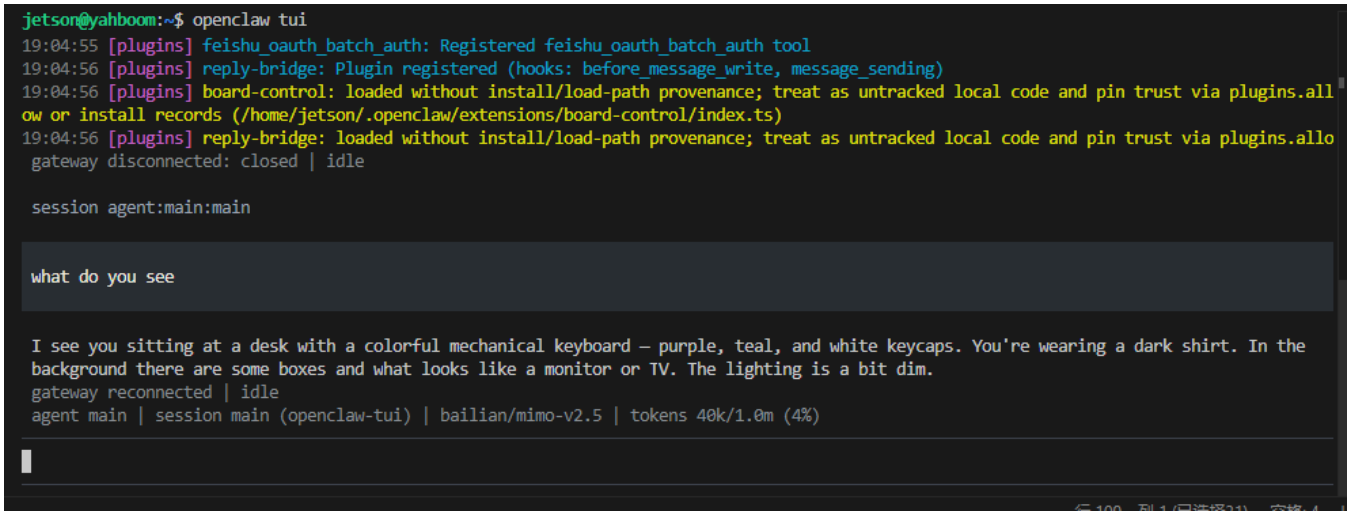
4. TUI Dialogue Configuration and Testing

4.1 Startup Method

Terminal input:

```
openclaw tui
```

If startup is successful, you will enter the local terminal dialogue interface.



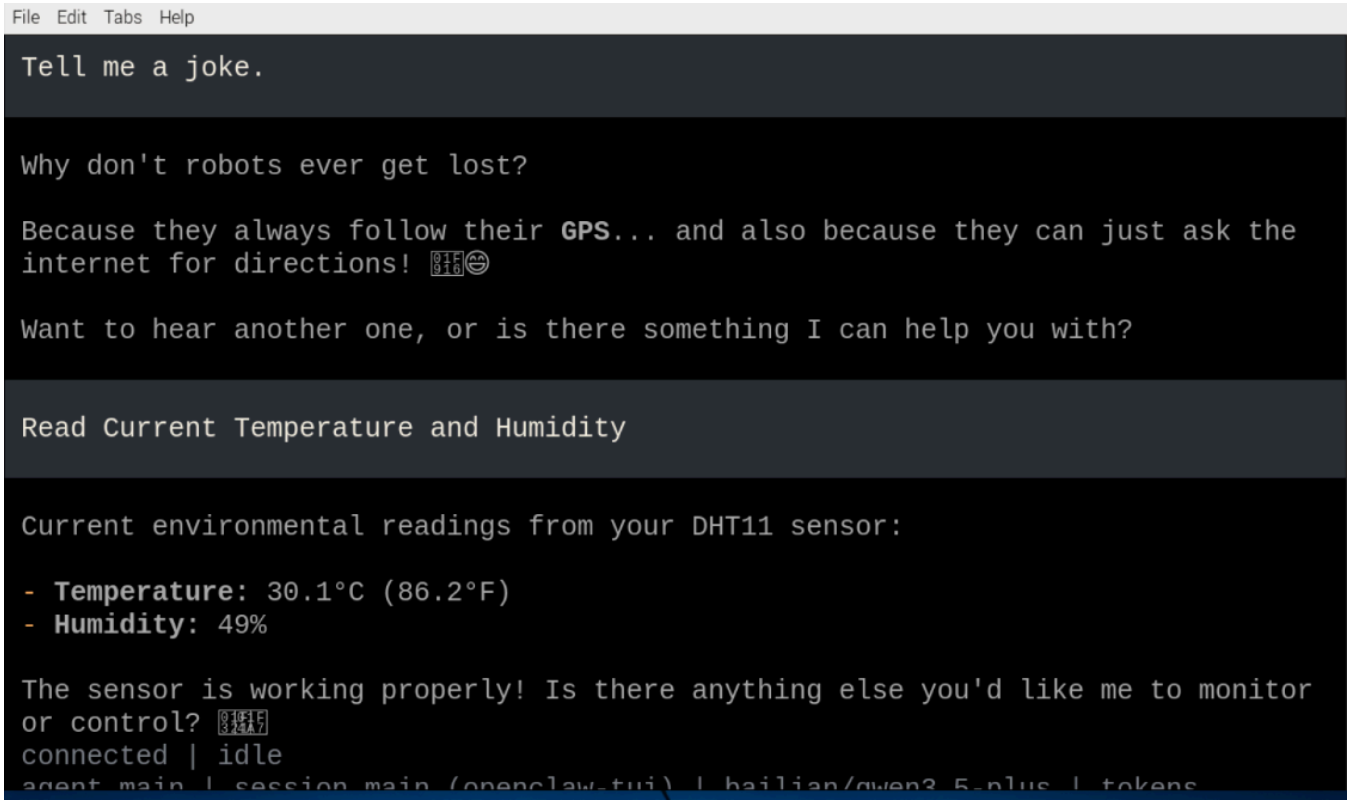
4.2 Testing Method

It is recommended to first input the simplest dialogue content:

- Tell me a joke
- Introduce yourself

If basic dialogue is normal, then test a hardware-related command, for example:

- Turn RGB light to red
- Display hello on OLED
- Read current temperature and humidity



5. AI Voice Interaction Configuration and Testing

5.1 Pre-startup Preparation

- Requires optional AI voice interaction module + speaker

5.2 Startup Method

```
cd /home/jetson/voice_to_openclaw
source .venv/bin/activate
python -m src.main
```

5.3 Testing Method

If using serial wake mode, first trigger wake-up with "Hello, Xiaoya".

Then you can try saying:

- Tell me a joke
- Turn on RGB light
- Display "Welcome to OpenClaw" on OLED

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.main
2026-05-13 20:00:29 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串
口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说"清空上下文"可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 2820 ms 语音, 正在转写

[你] Tell me a joke.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Why did the robot go to therapy?

Because it had too many *bugs* in
          its code! 🐛

...Don't worry, I'm working through my issues one debug
          at a time. 😄
[信息] 回复已裁剪为播报文本: 149 -> 90 字
```

5.4 Judgment Criteria

If voice method can complete recognition and receive OpenClaw reply, it indicates the following items are probably normal:

- Microphone collection is normal
- ASR is normal
- OpenClaw publishing pipeline is normal
- `[Optional]` TTS is normal

5.5 Common Problems

- No response to wake-up: First check serial wake configuration
- Can wake up but cannot recognize: Check ASR and API-KEY
- Can recognize but does not execute: Check OpenClaw gateway configuration
- Has return but no broadcast: Check TTS switch and speaker

6. Three Solutions Comparison

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote control, classroom demonstrations	Convenient for mobile and multi-person experience	Demonstrate "remote natural language control"

Solution	Suitable Scenarios	Advantages	Recommended Positioning
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	Preferred basic debugging solution
Voice interaction	Voice wake-up, complete experience	Strongest sense of interaction	Final demonstration solution

7. Recommended Testing Order

It is recommended to complete the three dialogue solution tests in the following order:

1. First test `openclaw tui`
2. Then test whatsapp robot
3. Finally test AI voice interaction